Delegate & Review Lab

Assignment 4.1 · Agentic AI & Collaborating with AI on Software Teams

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Dr. Paige Anne Rodeghero · Repository: `week4-delegate-review-lab`

Setup used: agentic, not simulated. Claude Code (terminal/CLI agent), with a supervising session on `claude-opus-5` driving a **delegated sub-agent** on `claude-sonnet-5` as the worker. The sub-agent had file read/write and shell access and retained its own context across all three checkpoints, so each gate resumed the same worker rather than briefing a fresh one. Between gates it planned, edited, and ran its own tests with no human prompt per step.

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Result in brief

The agent's work landed in two reviewed commits after three gates and one round of requested changes. 108 tests pass at the merge commit: 5 baseline tests unmodified since `93335ad`, 92 written by the agent, and 11 acceptance tests of mine. Ten of those eleven were committed at `ed6ad9f` before the agent received the task and were hidden from its branch; the eleventh was added at CP3, when AC8 was split in two. Valid-run output is byte-identical to the pre-change baseline.

The three most useful findings were mine, not the agent's. A latent `ZeroDivisionError` in my own baseline. An acceptance criterion (AC6) that tested zero *recipients* and never zero *opens*, three lines from the property that divides by opens. And AC8, which failed against **correct** code because it encoded a

specification my own CP1 ruling had superseded, in the very same edit where I amended AC6 for exactly that reason. Section 2 records all three.

Task Definition, Acceptance Criteria, and Checkpoints

Written and committed before the agent was given any instruction. The commit that adds this file is the parent of nothing on the agent's branch: the branch `feature/validation-and-top-n` is cut from the baseline commit `93335ad`, which means the agent's working tree never contained this document or the acceptance tests that ship with it.

- Course: CPSC 6820, Week 4 Assignment 4.1
- Baseline commit: `93335ad` (`campaign_report.py`, 198 lines, 5 tests passing)
- Agent branch: `feature/validation-and-top-n`
- Setup: agentic, not simulated. Claude Code (terminal/CLI agent) with a supervising session on `claude-opus-5` driving a **delegated sub-agent** on `claude-sonnet-5` as the worker. The sub-agent held file read/write and shell access and kept its own context across all three checkpoints, so each gate resumed the same worker rather than briefing a fresh one. Between gates it planned, edited, and ran its own tests with no human prompt per step.

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1. Task description given to the AI

Verbatim prompt in `docs/appendix-transcript.md`, section CP1. Substance:

> Add robust input handling and one new capability to `campaign_report.py`.
>
> **Part 1 - input validation and error handling.** The tool currently dies with a raw traceback on bad input. Every failure should instead be a clear, actionable message on stderr plus a non-zero exit code. At minimum handle: a missing or unreadable CSV path; a CSV missing required columns; non-numeric values in numeric columns; negative numbers; a campaign with zero recipients; and an empty CSV (header row only). Name the offending row and column where that makes sense.
>
> **Part 2 - new capability `--top N`.** Show only the N best campaigns by revenue per recipient, composing with the existing table and summary output.
>
> **Constraints.** Python 3, standard library only, no new dependencies. Do not change existing public function names, and do not change the output of a valid run. The 5 existing tests must keep passing. Add unit tests for new behavior under `tests/`. Keep it proportionate: this is a ~200 line script, so no package restructuring, no config system, no rewrite of the formatting layer.

The final constraint is a deliberate **scope-creep tripwire**. The module names scope creep as a known agent failure mode, so the task states an explicit proportionality bound and I recorded whether the agent honored it.

2. Acceptance criteria

Binary and observable. AC2 through AC7 and AC9 each additionally require that stderr contain no `Traceback`, because "it errors" and "it errors usefully" are different results.

ID	CRITERION
AC1	A valid run's stdout is byte-identical to the pre-change golden output in `tests/golden/baseline_report.txt`.
AC2	Missing/unreadable CSV path: message names the path, exit code non-zero.
AC3	CSV missing a required column: message names the missing column(s).
AC4	Non-numeric value in a numeric column: message names the row number and the column.
AC5	Negative value in a numeric column: rejected, row and column named.
AC6	A campaign with `recipients = 0`: skipped, the rest of the file still reported, and the skip announced on stderr. Never a `ZeroDivisionError`. *(Tightened at CP1, see note.)*
AC7	Header-only CSV: clear "no campaigns" message, exit code non-zero.
AC8	`--top 3` prints exactly 3 campaign rows: the 3 highest revenue-per-recipient, displayed in `--sort` order. *(Corrected at CP3, see note. Originally demanded descending RPR order, which my own CP1 ruling superseded.)*
AC9	`--top 0` and `--top -1` are rejected as invalid input.
AC10	All 5 baseline tests still pass, and the module imports nothing outside the standard library.

Note on AC8, and why it is a harder call than AC6. AC8 originally required `--top 3` to print the three campaigns in descending revenue-per-recipient order. It was the one acceptance test that failed against the agent's branch, and the failure was mine: at CP1 I ruled that `--top` selects while `--sort` orders, which contradicts AC8 as written. The agent implemented the ruling correctly. I amended AC6 for precisely this reason in the same commit, `a8dc4cd`, and never noticed AC8 needed identical treatment.

The AC6 amendment was easy to justify because it happened before any code existed. This one is not, and I want to be explicit about that rather than quietly fix it. I amended AC8 **after** seeing the implementation, which is the exact direction test gaming runs in, and "the code disagreed with my test so I changed my test" is the sentence a grader should be suspicious of.

What makes it refinement rather than gaming is that the superseding decision is timestamped, in writing, before the code: commit `a8dc4cd` records the `--top` filters / `--sort` orders ruling, and `50959d0` is where `--top` first exists. The amendment conforms the test to a documented earlier decision, not to whatever the agent happened to produce. Anyone can verify that ordering with `git log`.

The honest lesson is not that the amendment was fine. It is that my process had a hole: I refined one acceptance criterion at a checkpoint and failed to re-read the other nine against the same ruling. Had `--top` been genuinely wrong in the same way AC8 was stale, I would have had a failing test I was already primed to explain away.

Note on AC6. As first written, AC6 accepted either a hard failure or a safe render. At Checkpoint 1 I chose skip-and-continue over the agent's proposed hard failure, and tightened AC6 to require that the

skip be announced on stderr. The amendment was made before the agent wrote any code and while the acceptance tests were still outside its working tree.

This is worth distinguishing carefully, because it looks superficially like the failure mode it is the opposite of. Editing a test to match a decision the human made at a checkpoint is specification refinement, which is what checkpoints are for. Editing a test to match code the agent had already written would be test gaming. The ordering is what separates them, and the git history records the ordering: this amendment is commit 3, and the agent's first line of code lands after it.

3. My own test cases

Ten tests in `tests/test_acceptance.py`, written by me at this commit, covering AC1-AC10. An eleventh was added at CP3 when AC8 was split, noted below. Two representative ones, both required by the assignment:

Test case 1 (AC4) - non-numeric value names row and column. Feed a CSV whose `recipients` cell on data row 2 is the string `forty-thousand`. Require a non-zero exit, no traceback, and stderr that mentions both the row number and the column name. This is the case a naive `try/except Exception: print("bad CSV")` fix passes shallowly and fails on specificity, which is exactly the distinction I wanted to test.

Test case 2 (AC8) - `--top 3` selects by revenue per recipient. Against the committed fixture the correct three are Last Chance Spring Sale ($0.85), Bundle + Save 20% ($0.76), and Mothers Day Gift Guide ($0.68). Require exactly three campaign rows and that set of three. Per the CP1 ruling, display order is `--sort` order, which is date by default, so a companion test pins the descending-RPR order under `--top 3 --sort rpr`. Guards against both an off-by-one row count and ranking by the wrong column, since raw revenue would select a different third campaign.

As first written, this test additionally demanded descending RPR order in the default view. That is the stale requirement described in the note on AC8 above, and it is the one acceptance test that failed against correct code.

Why these are black-box subprocess tests. Every acceptance test shells out to `python campaign_report.py ...` and asserts on exit code, stdout, and stderr. The module's failure mode here is test gaming, and an agent can always satisfy an internal unit test by changing the internals it asserts against. It cannot satisfy an observable-behavior test without the behavior actually being correct.

Why they live only on `main`. Part A.3 of the module says to keep verification the agent cannot grade itself on. Committing these to `main` and cutting the agent's branch from the earlier baseline commit makes that structural rather than a matter of the agent's cooperation: the files are not in its working tree, so "please don't edit the tests" is not a rule it has the opportunity to break. I merge them onto the branch myself at Checkpoint 3.

4. Checkpoints

Three gates. The agent runs freely between them.

Checkpoint 1 - plan approval, before any code is written

Gate: the agent states its plan. No file edits until I approve. **I check:** does the plan cover all six error classes; does it invent structure I did not ask for; where does it put validation; does it plan to touch the existing tests or output format; does it treat `--top` as filter-then-summarize or filter-then-recompute (a real ambiguity in my spec, and I want to see whether it notices or silently picks). **Why here:** cheapest possible gate. Reviewing five lines of plan beats reviewing a 300-line diff, and per the module most bad outcomes are already visible in the plan.

Checkpoint 2 - mid-implementation, after validation, before `--top N`

Gate: stop when Part 1 is done and its tests pass. Do not start Part 2. **I check:** the actual validation diff, and specifically whether the baseline tests still pass without having been edited. `git diff` on `tests/` at this point is my test-gaming tripwire. **Why here:** this is the natural seam in the task, and it splits one large diff into two reviewable ones. It also puts a human between a possibly-wrong foundation and the feature built on top of it.

Checkpoint 3 - full diff review before merge

Gate: nothing merges to `main` until this passes. **I do:** read the entire diff line by line, not the agent's summary of it; copy my acceptance tests onto the branch and run them; run the baseline suite; write a real code review; only then restructure into clean commits and merge. **Why here:** the point of no return. `main` is the blast-radius boundary, so this is the one gate that cannot be skipped even if the first two look clean.

Guarded actions, autonomous actions. Reading files, editing files on the branch, and running tests are autonomous. Committing, anything touching `main`, `git push`, deleting files, and installing dependencies require me. The agent was told this.

Checkpoint Records

Setup: Claude Code (CLI agent) driving a delegated sub-agent session with file read/write and shell access. The sub-agent planned, edited, and ran its own tests between my gates without a prompt per step, which is the plan-act-observe loop from Part A.1. Full prompts and responses in `docs/appendix-transcript.md`.

Checkpoint 1 - Plan approval, before any code

Gate held. The agent was told to read the code but not to create, edit, or delete anything, and to answer with a plan only. It complied: it read four files across eight tool calls, confirmed the working tree was clean and 5/5 tests passed, and closed with "No files have been created, edited, or deleted." I verified that independently with `git status` rather than taking its word for it.

What it proposed

Verbatim in the appendix. In outline: one `CampaignReportError` exception; a `NUMERIC_FIELDS` type map to replace five repeated hardcoded casts; a `_parse_numeric_field(raw, column, row_number, cast)` helper doing conversion and the negative check together; all validation inside `load_campaigns` because that is the only place with row context; a `positive_int` argparse type for `--top`; and `build_report(campaigns, sort_key="date", top=None)` where `--top` selects which campaigns qualify and `--sort` still governs display order.

It also proposed two *new* test files rather than editing the baseline suite.

What I approved as-is

- **Validation inside `load_campaigns`.** Correct call. It is the only function that has both the file handle and the row index, so putting validation anywhere else would mean passing row numbers around for no reason.
- **The `NUMERIC_FIELDS` map.** This deletes a five-times-repeated cast. Fewer places for the next person to forget a check.
- **Two new test files instead of editing `tests/test_campaign_report.py`.** I had not asked for this. The agent's stated reason was that the baseline file's docstring says "happy path only" and editing it would make that false. That is a better instinct than I expected, and it is the same instinct that protects a baseline suite from being quietly reshaped to fit new code.
- **`--top` filters, `--sort` orders.** The two flags composing beats one overriding the other, and it is the reading that makes both flags still mean what their names say.

What I corrected

Correction 1 - rejected the blanket `except Exception` in `main()`. The agent proposed a broad catch-all as a "last-resort safety net so no code path can ever leak a raw traceback." I refused it. It optimizes for the letter of my requirement (no tracebacks) against its purpose (failures should be intelligible). A bare `except Exception` converts every genuine defect, including mine, into a polite one-line message and a silent exit 1, which is precisely how a real bug survives to production unnoticed. Catch `CampaignReportError` only; anything unanticipated should crash loudly with a full traceback. To its credit the agent flagged this itself and offered to drop it, which is the behavior I want at a plan gate: surface the arguable choice instead of burying it in a diff.

Correction 2 - message wording. It planned messages reading `"Row 4: ..."` while numbering 1-indexed *data* rows, so "Row 4" means the fifth line of the file. Anyone debugging will open the file and jump to line 4. I required the literal phrase `data row N`.

Correction 3 - bounded the doc edits. It wanted to add a `--top` example to the module docstring and to `README.md`. Approved, because documenting a flag is part of shipping it, but explicitly bounded: those two additions and nothing else in the README. Naming the bound was cheap, and left me a clean line to check the diff against later.

Its five questions, and my rulings

It surfaced five ambiguities unprompted. One of them, whether the summary block recomputes under `--top`, was an ambiguity I had deliberately left in the spec to see whether it would ask or silently pick. It asked.

1. **Does the summary recompute over the top-N subset?** Yes, recompute. A summary block that aggregates campaigns not listed in the table above it cannot be reconciled by the person reading it.
2. **Argparse-native rejection of `--top 0`, or route through `CampaignReportError`?** Argparse-native, exit code 2. Consistent with how `--sort` already validates, and a malformed flag is a usage error, not a data error.
3. **Scope of "unreadable" path.** Not-found, permission-denied, and is-a-directory under one `OSError` catch is sufficient.
4. **Row numbering convention.** 1-indexed data rows, but say `data row N`. See Correction 2.
5. **Zero recipients: hard failure or skip-and-continue?** It planned a hard failure for the whole run. **I overrode it: skip the row, report the rest, and print a warning naming the skipped campaign.** A single malformed row should not deny an analyst the other seven campaigns. But a silent skip is worse than a crash, because a report that quietly describes less than the file contains is how a wrong number reaches a client deck. So: continue, and say so loudly.

Did the checkpoint catch anything? Yes, and this is the part I would not have predicted. Nothing in the plan was *broken*. What the gate caught was two design choices I disagreed with (the catch-all, the hard failure) that would have been much more expensive to argue about after they were load-bearing in a 300-line diff, plus a wording problem that would have shipped. The module's claim that reviewing a five-

line plan beats reviewing a 500-line diff held up, though not for the reason I assumed. I expected the gate to catch errors. It caught *decisions*.

I amended AC6 to match my ruling on question 5 before releasing the agent to write code. See the note in `docs/01-task-definition.md` on why that is specification refinement and not test gaming.

Checkpoint 2 - Mid-implementation, after validation, before `--top N`

Gate held. The instruction was to finish Part 1 and stop, with no `--top` code, "not even the argparse wiring." It stopped. `parse_args` is untouched in the diff and the string `--top` does not appear anywhere on the branch.

Between my gates the agent worked autonomously across 17 tool calls: edited the module, created a test file of 23 functions, ran the suite, and iterated until green with no prompt from me between steps. That is the plan-act-observe loop, and it is the stretch of this exercise where I genuinely was not in the room.

What I verified myself instead of believing

Its self-report made four checkable claims. The module warns that agents summarize their own work optimistically, so I checked all four before reading any of the prose.

CLAIM	HOW I CHECKED	RESULT
Baseline tests untouched	`git diff --stat -- tests/test_campaign_report.py`	Empty. True.
33 tests pass	Ran the suite myself	33 passed. True.
Valid-run output unchanged	Diffed live stdout against my hidden golden file	Byte-identical, stderr empty, exit 0. True.
The `ZeroDivisionError` pre-exists in my baseline	Ran the same CSV against `git show 93335ad:campaign_report.py`	Identical traceback from the baseline. **True.**

All four held. I want that recorded plainly, because the honest lesson of CP2 is not the one I expected to be writing: at this gate the checkpoint's value was **confirmation, not detection.** That is not the same as the gate being useless. I could not have known the report was accurate without checking, and the fourth claim is precisely the kind an agent has an incentive to get wrong, since it assigns a bug to me rather than to itself.

The finding I did not expect: it found a bug in my code, through a hole in my own tests

The agent reported a reachable crash it deliberately did **not** fix. A CSV whose campaigns sum to zero opens or zero orders passes every validation check it added, then dies with a raw `ZeroDivisionError` in `format_summary` on `click_to_open_rate`. I verified it against the baseline commit: **it is my bug**, written into the baseline before the agent existed.

Worse for me, it is a bug my own acceptance criteria could not have caught. AC6 tests zero *recipients*. It never occurred to me to test zero *opens*, even though `click_to_open_rate` divides by `opens` three lines below the property I did think about. The agent found by reading what I failed to specify by writing.

That inverts the framing I began the lab with. I designed these checkpoints on the assumption that the human supplies correctness and the agent supplies speed. Here the agent supplied a correctness finding about the supervisor's own work, and the carefully pre-committed test suite was the artifact with the gap in it.

Scope: one unplanned file, self-disclosed

It created `tests/expected_report.txt`, a golden copy of the valid-run output, and flagged it unprompted as "not in my approved plan," offering to delete it.

That is scope creep in the literal sense. It is also the most interesting artifact of the lab, because **it duplicates a file I had already written and deliberately hidden from it.** My `tests/golden/baseline_report.txt` does the same job. The agent independently concluded that "do not change the output of a valid run" needed a golden-file guard and built one, because I had concealed the one that already existed.

So the isolation strategy carries a cost the module does not mention. Keeping verification outside the agent's control also keeps it outside the agent's *knowledge*, and an agent that cannot see your tests will rebuild them. I paid for that protection in duplicated work. I would still make the same trade, but I would now expect the duplication rather than be surprised by it.

I let the file stand through Part 2, where it actively guards output invariance, rather than remove a working guard mid-task. It was consolidated after the merge, in `9a05ddb`.

Rulings issued at this gate

1. **The `ZeroDivisionError`: leave it.** Its preferred option, rendering `n/a`, means inventing a display convention and editing `format_summary`, which I had explicitly fenced off. A defect that predates the branch and is unrelated to the delegated task does not belong in this change. On a real team that is a separate ticket, not a drive-by fix inside someone else's pull request. It becomes review comment 1 and a documented known issue instead.
2. **Helpers stay public.** Its reasoning beat my plan: the module has no underscore convention, so introducing one for four functions would add a second convention to a 200-line file. Approved as built.
3. **`tests/expected_report.txt` stays through Part 2**, and is consolidated after the merge in `9a05ddb`.
4. **Fix the BOM handling.** It noted a UTF-8 BOM would make the tool report `campaign_id` as a missing column. A CSV export carrying a BOM is not hypothetical, and this produces a *misleading* message from code written in this very change, so it is in scope. `encoding="utf-8-sig"` reads plain UTF-8 identically.

5. **Reword the integer message.** `41250.5` in `recipients` reported as "non-numeric," which is false. A misleading error message is the exact defect Part 1 exists to remove, so this is in scope despite being only wording.

I also audited its self-written tests for gaming before approving: 23 test functions collecting 28 cases through parametrization, no tautological assertions, and 50 assertions of which 48 pin actual message or output content. One of them, `test_unexpected_errors_are_not_swallowed`, is a test that enforces my own Correction 1 against future edits. I did not ask for that.

Part 1 was committed at this reviewed seam rather than left to accumulate into one large diff, which is why the branch history has two commits that match the two gates.

Checkpoint 3 - Full diff review before merge

Gate held, and this is the gate that earned its place. Outcome: **changes requested**, six items, then approved after remediation. Nothing reached `main` until the second pass.

Procedure, in this order deliberately: read the diff line by line *before* reading the agent's summary of it; run my held-out acceptance suite; reproduce every defect claim rather than accept it; hunt for defects the agent did not disclose; then write the review.

The acceptance suite: 9 of 10, and the failure was mine

I copied `tests/test_acceptance.py` and `tests/golden/` from `main` onto the branch for the first time and ran them. The agent's own 92 tests passed, and the branch suite went 97 for 97 counting my 5 baseline tests. My acceptance suite went 9 for 10.

AC8 failed against correct code. It demanded `--top 3` print the three campaigns in descending revenue-per-recipient order. The actual output selects the correct three (RPR 0.85, 0.76, 0.68 of eight) and displays them in date order. Which is exactly what I ruled at CP1: `--top` selects, `--sort` orders.

So the agent implemented my ruling, and my acceptance test still encoded the specification that ruling replaced. I amended AC6 for precisely this reason in commit `a8dc4cd`, and in the same edit failed to re-read the other nine criteria against the ruling I had just issued.

This is the most useful thing the lab produced, and not because the tests caught the agent. It is that the isolation worked in an unplanned direction. Holding the tests out of the agent's reach forced a real confrontation between my written spec and my later verbal ruling, and the confrontation exposed the drift. Had the acceptance tests been visible, the agent would most likely have reconciled the contradiction silently by satisfying whichever it read last, and I would never have learned I had contradicted myself. See `docs/01-task-definition.md` for why amending AC8 after seeing the code is defensible here, and where that line is genuinely thin.

Verifying the disclosed defect, and reversing my CP2 ruling

The agent reported that `--top` widened the reachability of the zero-denominator crash I ruled out of scope at CP2. I reproduced it on a two-row file whose highest-RPR campaign has revenue but no orders, an attribution-lag artifact rather than a contrived input:

```
python3 campaign_report.py no_orders.csv          -> clean report, AOV $100.00, exit 0
python3 campaign_report.py no_orders.csv --top 1  -> ZeroDivisionError, exit 1
```

I reversed my CP2 ruling. It was defensible when I made it and wrong by CP3, because the facts moved underneath it: the division is old, but which inputs reach it is new, and this change is what made them reach it. A change that turns a working command into a traceback on the same file has introduced a regression regardless of who authored the arithmetic. "Pre-existing" describes a line of code; it does not describe the behavior, and the behavior is what regressed. I unfenced `format_summary` for the fix, since that fence existed to prevent gratuitous rewriting, not to compel shipping a crash.

What I found that the agent did not disclose

`build_report(campaigns, top=0)` raised `ZeroDivisionError`. `positive_int` guards the CLI, but `top_campaigns` and `build_report` are public and an empty selection reached `totals()`. I found it by deliberately looking past the agent's list of its own defects rather than through it, which became the practical lesson of this gate: a candid self-report is more useful than a silent one and also more seductive, because a thorough-sounding enumeration invites you to treat it as complete.

I also reproduced the agent's `SORT_KEYS` coupling concern and found it worse than described. Reassigning `SORT_KEYS["rpr"]` to ascending, a plausible display-only edit, silently made `--top 3` return the three *worst* campaigns.

The agent corrected my reasoning, and was right

On remediation it pushed back on one of my six items. I had justified the empty-selection guard as preventing a division by zero. It pointed out that once `ratio()` returns `None` for a zero denominator, `build_report(campaigns, top=0)` no longer divides by zero at all: it returns a header with no rows and a summary reading `0 campaigns | 0 recipients | $0.00 revenue`. The guard therefore earns its place by refusing to describe nothing, not by preventing a crash.

I verified this rather than take it. `ratio(5, 0)` returns `None` and `totals([]).recipients` is 0. There are two guards, not one: `top_campaigns` raises `ValueError: count must be 1 or more, got 0`, which is the one `top=0` actually reaches, and `build_report` raises `ValueError: cannot build a report from zero campaigns` for an empty list. Its correction stands and my stated rationale in the review was wrong. Recorded because the two rulings interact, and a review record should not preserve a justification I know to be incorrect.

It also declined to route that guard through `CampaignReportError`, reserving that type for user-fixable input and using `ValueError` for a programming error. I had not specified which. That is the right distinction and I did not ask for it.

Remediation verified

97 tests pass (5 baseline untouched, 29 validation, 40 `--top`, 23 formatting). `git diff 93335ad -- tests/test_campaign_report.py` is still empty. Valid-run output is byte-identical to the baseline golden file with stderr empty. The formerly-crashing invocation now exits 0 and degrades only the genuinely undefined metric, printing `AOV n/a` while `CTOR`, `RPR`, and `CVR 0.0%` stay real numbers, since zero orders over a real denominator is a defined rate of zero rather than an undefined one.

The agent also disclosed that two of its own tests failed during remediation and that it changed the tests rather than the code, because the pluralization fix I requested is what made `"1 campaigns"` wrong. Correct call, disclosed without being asked.

Merged with `--no-ff` after this pass. Full review in `docs/03-code-review.md`.

Code Review

Reviewing: `feature/validation-and-top-n` as presented for merge at Checkpoint 3, being Part 1 (commit `0032474`) plus the Part 2 working tree. **Author:** Claude Code (`claude-sonnet-5`), delegated agent. **Reviewer:** Jeff Branyon. **Verdict:** Changes requested, then approved after remediation.

I reviewed the diff rather than the agent's summary of the diff, and I re-derived every factual claim in its self-report instead of accepting it. Where a comment below asserts a defect that is reachable today, I reproduced it first, and the reproduction is included so the claim is checkable rather than trusted. Comment 4 is the exception and says so: it describes a latent fault that cannot currently be triggered. Comments are ordered by what I would want fixed first, not by where they appear in the file.

Two notes on calibration before the comments. First, the module warns that AI-generated code "looks more polished than its correctness warrants," and that is a fair description of this diff: it is well-organized, the docstrings are accurate, and the naming is consistent, which made it pleasant to read and therefore easy to under-scrutinize. Second, this diff arrived with an unusually candid self-report, which changed my job in a way I did not anticipate. Several of the defects below were disclosed to me by the author. That is genuinely useful, and it is also a subtler review hazard than an agent that hides things: a thorough-sounding self-assessment invites you to treat its list as the complete list. Comment 3 is one I found only because I went looking past it.

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1. BLOCKING, correctness. `--top` widens the reachability of a zero-denominator crash

Where: `build_report` / `format_summary` interaction.

Because the summary now recomputes over the selected subset, a subset can have a zero denominator when the whole file does not. This is a two-row file whose highest-RPR campaign has revenue but no orders, which is an ordinary attribution lag artifact, not a contrived input:

```
$ python3 campaign_report.py no_orders.csv          # clean report, AOV $100.00, exit 0
$ python3 campaign_report.py no_orders.csv --top 1  # ZeroDivisionError traceback, exit 1
```

The author flagged this and correctly noted that my earlier ruling no longer covered it. I agree, and I am reversing that ruling rather than defending it.

At Checkpoint 2 I ruled this out of scope as a pre-existing defect deserving its own ticket. That reasoning was sound at the time and is wrong now, because the facts changed underneath it. The division itself is old, but *which inputs reach it* is new, and this change is what made those inputs reach it. A change that turns a working command into a traceback on the same file has introduced a regression, whether or not it authored the arithmetic. "Pre-existing" describes the line of code, not the user-visible behavior, and the user-visible behavior is what regressed.

Requested: render `n/a` for click-to-open rate and average order value when the denominator is zero. I explicitly unfenced the formatting layer for this, since that fence existed to prevent gratuitous rewriting, not to force shipping a crash.

Resolved. A `ratio()` helper returns `None` on a zero denominator, all six rate properties route through it, and `pct()` / `money()` render `None` as `n/a`. The formerly-crashing command now exits 0. Critically, only the genuinely undefined metric degrades: `AOV n/a` while `CTOR`, `RPR`, and `CVR 0.0%` remain real numbers, because zero orders over a real denominator is a defined rate of zero, not an undefined one. That distinction is pinned by two tests, one of which is named `test_ratio_does_not_report_undefined_as_zero`. Conflating the two would have been the easy wrong fix, and it is the version I would have written.

2. BLOCKING, maintainability. Selection semantics are borrowed from a display constant

Where: `top_campaigns`.

The line under review, which is not what `main` contains today:

```
return sorted(campaigns, key=SORT_KEYS["rpr"])[:count]
```

`SORT_KEYS` exists to order rows for display. Reusing its `rpr` entry to decide *which campaigns qualify* couples a correctness guarantee to a presentation detail, and nothing in either location says so. The author raised this as a DRY-versus-safety tradeoff. It is worse than that framing suggests, and I verified how much worse:

```
# Against the branch AS REVIEWED, before this comment was addressed:
cr.SORT_KEYS['rpr'] = lambda x: x.revenue_per_recipient   # a display-only change
# before the flip -> Last Chance Spring Sale, Bundle + Save 20%, Mothers Day Gift Guide
# after the flip  -> Founder Story, New Scent Announcement, Bestsellers Restock
```

That snippet no longer reproduces on `main`, which is the point of the fix. Selection now reads `revenue_per_recipient` directly, so flipping the display constant leaves `--top` unchanged.

It is also not recoverable from the history, and that is worth stating rather than leaving for someone to discover. The coupled version was never committed. I reviewed the working tree and requested the fix before Part 2 was committed, so `50959d0` already contains the corrected `top_campaigns`. Committing only at reviewed seams keeps the history clean and costs you the ability to diff against the rejected state. The reproduction above is the record of it.

Someone changing a sort direction for display reasons, with no reason to think they are touching `--top`, silently inverts the feature. `--top 3` then returns the three *worst* campaigns, in a tool whose output goes into client reporting. The failure is silent, plausible, and lands on data rather than on an exception.

This is the comment I would push hardest on even though nothing is broken today, because the cost of being wrong is asymmetric. The bug it prevents is invisible, and the fix is three lines.

Requested: select on `revenue_per_recipient` directly, and add a test that fails if the display lambda is flipped.

Resolved, and the agent went one better on verification. Selection now sorts on the property directly. It then confirmed the new tests actually discriminate by temporarily reverting `top_campaigns` to the `SORT_KEYS` version and checking that both decoupling tests fail against the old code and pass against the new. One of them, `test_top_selection_survives_removing_rpr_from_sort_keys`, additionally fails with `KeyError: 'rpr'` on the old implementation, so the pair catches a flipped constant *and* a deleted one. Writing a regression test is routine; proving it fails against the bug it targets is the step most people skip, including me.

3. BLOCKING, correctness. Validation lives only at the CLI boundary

Where: `positive_int` guards `--top`; `top_campaigns` and `build_report` do not.

Not disclosed in the self-report, and the one I found by looking past the author's list rather than through it.

```
>>> cr.build_report(cr.load_campaigns('data/campaigns.csv'), top=0)
ZeroDivisionError: division by zero
```

`positive_int` rejects `--top 0` at the command line, so the CLI is safe. But `build_report` and `top_campaigns` are public functions, an empty selection reaches `totals()`, and summing an empty list yields zero recipients. Guarding an invariant at the outermost boundary is fine right up until a second caller appears, and a second caller already exists: the test suite imports and calls these functions directly.

Requested: guard the selection path itself so no caller can drive it to a division by zero.

Resolved, and my rationale just above was wrong. The agent implemented the guard, then corrected me: once comment 1's fix makes `ratio()` return `None` for a zero denominator, `build_report(campaigns, top=0)` no longer divides by zero at all. It returns a header with no rows and a summary reading `0 campaigns  |  0 recipients  |  $0.00 revenue`. I verified that rather than take it. `ratio(5, 0)` is `None` and `totals([]).recipients` is 0, so the correction holds. The guard still belongs, but it earns its place by **refusing to describe nothing**, not by preventing a crash. There are now two guards. `top_campaigns` raises `ValueError: count must be 1 or more, got 0`, which is the one a `top=0` call actually reaches, and `build_report` raises `ValueError: cannot build a report from zero campaigns` for an empty list. The agent deliberately chose `ValueError` over `CampaignReportError` because the latter is documented as user-fixable input while `top=0` from a caller is a programming error. I had not specified which. That distinction is right and I did not ask for it.

I am leaving my incorrect reasoning in place above rather than quietly rewriting it, because a review record is more useful when it shows where the reviewer was wrong.

4. Should fix. The error reporter can raise while reporting an error

Where: `parse_numeric`, `NUMERIC_KINDS[cast]` inside the `except` block.

A cast added to `NUMERIC_COLUMNS` without a matching `NUMERIC_KINDS` entry makes the *reporter* fail, replacing a clean message with a `KeyError` traceback from inside exception handling. Unreachable today, since both casts are mapped, and I still want it changed: this whole change exists to replace tracebacks with intelligible messages, so a latent traceback in the message path is the one place that is thematically indefensible. The author left it deliberately, arguing a missing entry is a misconfiguration that should be loud. Reasonable, but it will be loud in the least informative possible way, and at the least convenient moment.

Requested: `NUMERIC_KINDS.get(cast, "a number")`.

Resolved, with a self-caught bad verification. The agent applied the fallback, then reported that its first attempt to verify it proved nothing: it tested with `Decimal`, whose `InvalidOperation` is an `ArithmeticError` and therefore escapes `except (TypeError, ValueError)` before any message gets built. It caught that itself, retested with an unmapped cast that raises `ValueError`, and got the correct fallback message. It then flagged the residual issue, that a future non-`int`/`float` cast could still escape the `except` pair entirely, and did not fix it because it was not among my six items. Correct on both counts. A test that passes for the wrong reason is worse than no test, and noticing that about your own verification is harder than writing the fix.

5. Question, not a defect. Should selection follow `--sort` instead of always ranking by RPR?

Where: `build_report`.

Selection is hardcoded to revenue per recipient regardless of `--sort`, so `--sort open --top 3` returns the top three by *revenue per recipient*, displayed in open-rate order. A user who asked to sort by open rate and take the top three may well have expected the top three by open rate.

I want to be clear that this is a question about my own specification, not about the implementation. The agent built precisely what I ruled at CP1, and the ruling is defensible: `--top` means "best campaigns," and a brand's definition of best is revenue per recipient, not open rate. But I only noticed the interaction reading the finished feature, which suggests I ruled on the composition of two flags without having thought through all ten combinations. If this tool acquires real users, the `--sort X --top N` combination is the first thing I would expect a bug report about.

Action: no change requested. Documented so the next person inherits the reasoning rather than rediscovering the surprise.

6. Nits

- **`positive_int` catches `TypeError`**, which argparse cannot hand it. Dead defensive code, carried over from `parse_numeric` where `None` genuinely can arrive from `DictReader`. Author self-

flagged. Requested: drop it.

- "**1 campaigns.**" Pre-existing in `format_summary`, but `--top 1` promotes it from an edge case reachable only via a one-row CSV to a normal command, and it now appears in real output. Requested, since the formatting layer was unfenced anyway.
- **`row_names` in `tests/test_top_n.py`** splits rows on a double space to extract campaign names, which breaks on a name containing two consecutive spaces. Author self-flagged it as the weakest thing in the new tests. Not requested: it is a test helper, the fixtures do not trigger it, and I would rather not spend a remediation round on it.

7. Process. An already-approved file was modified

Where: `tests/test_validation.py`, +19/-2, a file I had reviewed and committed at Checkpoint 2.

The changes themselves are correct and I would have asked for them: regression guards pinning the two rulings I issued at CP2, neither of which had a test. Without them, both rulings would have been one careless edit from silently reverting.

I am flagging it anyway, because the thing that makes it acceptable is the disclosure, not the diff. The agent volunteered it under a heading reading "Changed outside what you listed." An identical edit made silently would have been the more serious problem, since a file I have already signed off on is exactly where I am least likely to look twice. Modifying approved work is sometimes right; doing it quietly never is.

Done well

Recorded because a review that only lists defects misrepresents the change.

1. **It reported a bug as mine and let me check.** The `ZeroDivisionError` was in my baseline. It said so, cited `git show HEAD:campaign_report.py`, which on its branch at that point resolved to `93335ad`, as the evidence, and declined to fix it because the fix needed a decision I had not made. It would have been easier to quietly patch it, or to omit it. I verified the attribution before accepting it, precisely because it was the claim most convenient for the author.
2. **It reported its own wrong test expectation.** One test failed initially because it had eyeballed a ranking rather than computing it. It fixed the expectation, added the real values as a comment, and noted, unprompted, that "it would have been easy to fix the code to match my wrong expectation." That is the test-gaming failure mode named in the module, identified by the author in its own work, in the specific direction where it would have been hardest for me to notice.
3. **It wrote a test enforcing a correction against me.** `test_unexpected_errors_are_not_swallowed` pins my Correction 1, the rejection of a blanket `except Exception`. I did not ask for it. It protects my decision from my own future edits.
4. **It verified the provenance of its own fixture.** Rather than assert that the golden output file was the true baseline, it regenerated the original script from `93335ad` and diffed against it, which is the check that distinguishes a real invariance guarantee from one built after the fact.

One finding that is not about this code

My held-out acceptance suite passed 9 of 10 against this branch. The failure, AC8, was mine. AC8 demanded that `--top 3` print the three campaigns in descending RPR order; at CP1 I ruled that `--top` selects and `--sort` orders. The agent implemented the ruling. I never propagated the ruling into AC8, in the very same edit where I amended AC6 for the same reason.

Full accounting in the Checkpoint 3 record. It belongs in the review only to be explicit that the branch was not charged for it.

Resolution summary

#	COMMENT	CLASS	OUTCOME
1	`--top` widens a zero-denominator crash	Blocking, correctness	Fixed. `n/a` for undefined rates only
2	Selection borrowed from a display constant	**Blocking, maintainability**	Fixed. Decoupled, with tests proven to fail against the old code
3	Public API unguarded where the CLI is guarded	Blocking, correctness	Fixed. `ValueError` guard. My rationale was wrong and the agent corrected it
4	Error reporter can raise while reporting	Should fix, robustness	Fixed. Agent caught its own invalid verification
5	Should selection follow `--sort`?	Question, design	No change. Documented; it is a question about my spec, not the code
6	Dead `except`, "1 campaigns", weak test helper	Nits	First two fixed. Third accepted, not worth a round
7	An already-approved file was modified	Process, scope	Accepted. The disclosure is what makes it acceptable

Comment 2 is the beyond-correctness comment: nothing was broken, and I pushed hardest on it anyway, because the cost of being wrong is asymmetric and silent.

Approved and merged with `git merge --no-ff` after remediation. 97 tests pass, the 5 baseline tests remain untouched since `93335ad`, valid-run output is byte-identical, and my held-out acceptance suite passes 11 of 11 once AC8 was corrected to match the CP1 ruling it had contradicted.

Reflection

Where the checkpoints were, and what they caught. Three gates. Plan approval before any code. A stop after the validation layer. A full diff review before merge. Each caught something. None caught what I built them to catch.

CP1 caught decisions, not errors. The agent wanted a blanket `except Exception` in `main()` so no path could "leak a raw traceback." That meets my requirement and defeats its point, since every real bug becomes a polite exit 1. It also planned to fail the whole run on a zero-recipient row, where I wanted a skip and a warning. Both would have been harder to argue once buried in the finished diff. So the module is right that a five-line plan beats a 500-line diff, but not for the reason I expected. I looked for mistakes and found judgment calls.

CP2 confirmed rather than detected. The agent made four checkable claims and all four held up when I ran them myself. I could not have known that otherwise. And the fourth was the claim it had most reason to fudge. It blamed a bug on me instead of itself.

CP3 earned its place. The agent told me its new `--top` flag made an old crash reachable. Same CSV file: clean report without the flag, traceback with it. That reversed my CP2 ruling that the bug was out of scope. The line was old, but which inputs reached it was new.

Failure modes. Test gaming: none. It never touched the baseline tests. One of its own tests failed and it told me the test was wrong, not the code, noting unasked that a lazier fix would have edited the code to match. Scope creep: one small case, a golden output file outside its plan, which it flagged itself. Confident wrong turns: none. It asked instead, raising five gaps in my spec at CP1, including one I planted.

Here is what I did not expect. The failures I was watching for showed up in me. The `ZeroDivisionError` was in my own baseline. My AC6 tested zero recipients but never zero opens, though `click_to_open_rate` divides by opens three lines away. And AC8 failed against correct code, because it still held a spec my own CP1 ruling had replaced. I edited AC6 for that reason in the same commit and never re-read the other nine. I built this to catch the agent drifting. It caught me.

How reviewing it felt different. Cleaner, and more dangerous. The diff was better organized than a classmate's first draft and better documented than my baseline. That is the polish-beats-correctness problem the module warns about. But the bigger trap was how honest it was about its own bugs. A list that thorough makes you assume it is complete. It missed one: a public function unguarded while the CLI was guarded. I only found that by looking past its list. Reviewing a classmate, I would also be managing how they feel. Here there was nothing to manage. That sounds like a win. Mostly it means nothing makes you slow down when you should.

Would I trust this on a real team? Yes, with three conditions. Keep some tests where the agent cannot reach them, since that is why AC8 turned up. Put gates at points of no return, not on a timer, because the pre-merge gate caught the real problem and the mid-run gate did not. And whoever merges owns it,

because "the AI wrote it" will not survive an incident review. One cost I missed: the agent rebuilt a guard I had, since it could not see mine.

Version Control Record

The graph, the commit structure, and the attribution convention. Generated from the repository, not transcribed.

Branch strategy

`main` never received a direct commit from the agent. The agent worked only on `feature/validation-and-top-n`, and that branch was cut from the **baseline** commit `93335ad` rather than from the tip of `main`. That single choice is what made "keep verification outside the agent's control" structural instead of advisory: the task definition, the acceptance criteria, and the ten acceptance tests that existed at that point were committed to `main` at `ed6ad9f`, which is not an ancestor of the agent's branch, so those files were never present in its working tree. An eleventh test was added at CP3, also on `main` only. It could not read them and could not edit them.

The topology below shows this directly. The branch leaves history at `93335ad`, below every docs commit, and rejoins only at the merge.

The graph

The lab itself, from the baseline through the merge and the fixture consolidation that followed it. Every commit after `9a05ddb` is a linear tail of documentation and build commits, listed in the table below.

```
* 9a05ddb Consolidate duplicate golden output fixture
*   7f6408a Merge branch 'feature/validation-and-top-n'
|\
| * 50959d0 Add --top N ranking by revenue per recipient
| * 0032474 Validate CSV input and fail with actionable messages
* | b5ab724 Record CP3 review, correct AC8, add review and reflection
* | f85d623 Record CP2 review: verified claims, five rulings
* | a8dc4cd Record CP1 plan review; refine AC6 to skip-and-warn
* | ed6ad9f Define delegated task, acceptance criteria, and checkpoints
|/
* 93335ad Add campaign_report CLI baseline
```

Commits, and what each one is for

COMMIT	AUTHOR	PURPOSE
`93335ad`	Human	Baseline program. 198 lines, 5 tests. The thing being delegated against.
`ed6ad9f`	Human	Task, AC1-AC10, checkpoints, and the held-out acceptance tests. Committed **before** the agent was given any instruction, so the timestamp is the evidence that the checkpoints were designed in advance.
`a8dc4cd`	Human	CP1 record. Three corrections, five rulings, AC6 tightened after I overrode the agent's plan.
`0032474`	**Agent**, reviewed	Part 1: input validation with actionable errors. Committed by me at the CP2 seam, after review.
`f85d623`	Human	CP2 record. Four self-reported claims independently verified, five rulings issued.
`50959d0`	**Agent**, reviewed	Part 2: `--top N`, plus the `n/a` fix that landing it safely required.
`b5ab724`	Human	CP3 record, code review, reflection. AC8 corrected.
`7f6408a`	Merge	`--no-ff`, so the branch topology survives in the history.
`9a05ddb`	Human	Consolidates the duplicate golden fixture the isolation caused.
after `9a05ddb`	Human	Documentation and build commits only: this record, the submission PDFs, a fix to the audit query below, a revision of the reflection, folding the 6820 part into the same PDF, and correction passes over these documents. No program code changes after the merge.

Two agent commits, one merge, and the rest human. The agent's work landed in two commits because I committed at each reviewed checkpoint, rather than letting one large diff accumulate and then rewriting history to look tidy afterward. Both agent commits pass the module's test: each has an honest one-line summary.

Attribution convention

The two commits containing agent-authored code carry three trailers each. The merge that brought them to `main` carries the first two, since a merge introduces no newly authored code. Every other commit carries none.

```
Assisted-by: Claude Code (claude-sonnet-5), delegated agent, Part 1 of 2
Reviewed-by: Jeff Branyon <jeff@copyrath.com> at Checkpoint 2
Co-Authored-By: Claude <noreply@anthropic.com>
```

`Assisted-by` is the course convention and carries the audit detail: which model, in what role, and which slice of the task. `Reviewed-by` names the human who accepted it and the gate where that happened, which is the trailer that actually assigns ownership. `Co-Authored-By` is the machine-readable git-standard form, so the attribution is greppable by tooling that does not know about the other two.

The module says consistency is what makes attribution useful for later audit, so the test should be mechanical. **Anchor the pattern to the trailer line:**

```
$ git log --grep='^Assisted-by:' --oneline
7f6408a Merge branch 'feature/validation-and-top-n'
50959d0 Add --top N ranking by revenue per recipient
0032474 Validate CSV input and fail with actionable messages
```

Exactly the three commits carrying agent-authored code: the two agent commits and the merge that brought them to `main`. Every other commit in the repository is human-authored and carries no trailers. That set is final, since no further agent work followed the merge.

Why anchored, which I learned by getting it wrong. I first documented this as `git log --grep='Assisted-by'`, unanchored, and asserted it returned "exactly the commits containing agent-authored code, and nothing else." Then I wrote a commit message that *discussed* the convention, and the unanchored pattern matched its prose. The audit query started returning human-only commits, two of them by the end, and the claim in this document stopped reproducing about sixty seconds after I made it.

It is a small bug with a real point behind it. An attribution convention is only as good as the query that audits it, a substring match over a whole commit message is not that query, and I found this only because I re-ran my own documented command against the pushed repository instead of trusting the output I had already pasted in. `^Assisted-by:` matches the trailer and not discussion of the trailer.

Verification at the merge commit

```
$ python3 -m pytest tests/ -q
108 passed in 0.40s

$ git diff 93335ad --stat -- tests/test_campaign_report.py   # baseline suite untouched
(no output: the 5 baseline tests were never modified)

$ python3 campaign_report.py data/campaigns.csv | diff - tests/golden/baseline_report.txt
(no output: valid-run output is byte-identical to the baseline)
```

Appendix: Prompts and Transcript

Setup: Claude Code (CLI agent) as supervising session, driving a **delegated sub-agent** with file read/write and shell access. The sub-agent held its own context across all three checkpoints, so each gate resumed the same worker rather than briefing a fresh one. Between gates it planned, edited, and ran its own tests with no human prompt per step.

Reproduced verbatim. Tool-call chatter and file dumps are elided where marked, and nothing in the agent's prose is edited, including the parts where it is wrong.

One mechanical exception, noted so that "verbatim" stays true: dash characters are normalized throughout this document, with em and en dashes rendered as hyphens, in both my prompts and the agent's replies. No wording, claim, or figure is altered.

CP1 - Prompt: task delegation and plan gate

> You are acting as a delegated coding agent on a small Python repository. I am your human supervisor. I have defined checkpoints where you must stop and wait for me.
>
> Repository: /Users/jbranyon/Documents/CPSC/6820/week4-delegate-review-lab Current branch: feature/validation-and-top-n (already checked out. Work here, never on main.)
>
> YOUR TASK (two parts):
>
> Part 1 - input validation and error handling. `campaign_report.py` currently dies with a raw traceback on bad input. Every failure should instead be a clear, actionable message on stderr plus a non-zero exit code. At minimum handle: a missing or unreadable CSV path; a CSV missing required columns; non-numeric values in numeric columns; negative numbers; a campaign with zero recipients; and an empty CSV (header row only). Name the offending row and column where that makes sense.
>
> Part 2 - new capability `--top N`. Show only the N best campaigns by revenue per recipient, composing with the existing table and summary output.
>
> CONSTRAINTS:
>
> - Python 3, standard library only. No new dependencies.
> - Do not change existing public function names. Do not change the output of a valid run.
> - The 5 existing tests in tests/test_campaign_report.py must keep passing.
> - Add unit tests for new behavior under tests/.
> - Keep it proportionate. This is a ~200 line script: no package restructuring, no config system, no rewrite of the formatting layer.
>
> ACTION POLICY:
>
> - Autonomous: reading files, editing files on this branch, running tests.
> - Requires my explicit approval: any git commit, anything touching main, git push, deleting files, installing dependencies.

> CHECKPOINT 1 - PLAN APPROVAL. This is the gate you are at right now.
>
> Do NOT edit, create, or delete any file yet. Reading files is encouraged - read the code before you plan.
>
> Respond with your implementation plan only:
>
> 1. How you will structure validation and where it will live.
> 2. How you will handle each of the six error classes.
> 3. Your approach to --top N.
> 4. What tests you will add.
> 5. Any ambiguity in my spec you want me to resolve before you start.
>
> Then stop. I will approve or correct the plan before you write any code. Your response is the plan itself - it goes into my checkpoint record verbatim, so write it as a plan, not as a status update.

CP1 - Agent response (verbatim, 8 tool calls, all reads)

> ### Implementation Plan - `feature/validation-and-top-n`
>
> **Repo state confirmed before planning:** branch `feature/validation-and-top-n` already checked out, working tree clean, single commit `93335ad Add campaign_report CLI baseline`. `python3 -m pytest tests/ -q` currently passes 5/5. Files in scope: `campaign_report.py` (~200 lines), `tests/test_campaign_report.py` (5 baseline tests), `README.md`, `data/campaigns.csv` (8-row fixture). No `docs/` directory exists yet despite the README referencing one. No files have been created, edited, or deleted - this response is planning only.
>
> #### 1. Structure and location of validation
>
> Everything stays in `campaign_report.py` - no new module, no validation class hierarchy.
>
> - **New exception**: `class CampaignReportError(Exception): pass`, defined near the top (after `EXPECTED_COLUMNS`). This is the one "expected, user-facing" error type. `main()` catches it and nothing else needs to change shape.
> - **New field-type map**: `NUMERIC_FIELDS = {"recipients": int, "opens": int, "clicks": int, "orders": int, "revenue": float}` - replaces the five hardcoded `int(row[...])` / `float(row[...])` calls with one small loop, so the parse-and-check logic isn't repeated five times.
> - **New helper**: `_parse_numeric_field(raw, column, row_number, cast)` - does the `cast(raw)` conversion, catches `(TypeError, ValueError)` and raises `CampaignReportError` naming the row and column, then checks `< 0` and raises `CampaignReportError` for negatives the same way. Used for all five numeric columns uniformly.
> - **All validation lives inside `load_campaigns`**, since that's already where the file is opened and rows are iterated - it's the natural single place that has the row context needed to name offending rows. `load_campaigns(path)` keeps its exact existing signature and, for valid input, builds `Campaign` objects identically to today (same conversions, same order), so a valid run's output is provably unchanged.
> - `main()` gets a `try/except CampaignReportError as exc: print(f"Error: {exc}", file=sys.stderr); return 1`, plus one broader `except Exception as exc: ... return 1` as a last-resort safety net so no code path can ever leak a raw traceback (flagged as a design choice below - happy to

drop the broad catch-all if you'd rather see real tracebacks for truly unanticipated bugs during grading/debugging).

- All six handled error classes exit with code `1`. Argparse's own CLI-usage errors (bad `--sort` choice, bad `--top` value, missing positional arg) keep argparse's existing exit code `2`, unchanged.

2. Handling each error class

#	CASE	WHERE CAUGHT	MESSAGE SHAPE
1	Missing/unreadable CSV path	`open()` call in `load_campaigns`, wrapped in `try/except OSError`, branching on `FileNotFoundError` / `PermissionError` / `IsADirectoryError` / generic `OSError`	`"CSV file not found: '<path>'"` / `"Permission denied reading '<path>'"` / `"'<path>' is a directory, not a CSV file"` / generic fallback with the system message
2	Missing required column(s)	Right after `csv.DictReader(handle)`, check `reader.fieldnames` against `EXPECTED_COLUMNS` before looping rows	`"'<path>' is missing required column(s): orders, revenue"` (names all missing columns at once)
3	Non-numeric value in a numeric column	`_parse_numeric_field`, `except (TypeError, ValueError)`	`"Row 4: column 'clicks' has a non-numeric value 'N/A'"`
4	Negative number	`_parse_numeric_field`, post-cast `< 0` check	`"Row 4: column 'revenue' is negative (-12.50); expected a value >= 0"`
5	Zero recipients	Explicit check right after a row's `recipients` is parsed, before constructing the `Campaign`	`"Row 4: campaign 'c1004' has 0 recipients; cannot compute rate metrics"`
6	Empty CSV (header only)	After the row-reading loop finishes, `if not campaigns: raise ...`	`"'<path>' has no data rows (header only)"`

Row numbers are 1-indexed **data rows** (first row under the header = row 1), not raw file line numbers - flagged as a choice in section 5. Column/row naming is included wherever it's meaningful (cases 3-5); cases 1, 2, 6 are file- or column-level so they name the path/column instead, per "where that makes sense." As a small bonus (not a new category, just a clearer message for a true 0-byte file with no header at all), I'll special-case `reader.fieldnames is None` before the missing-columns check: `"'<path>' is empty (no header row found)"`.

3. `--top N` approach

- `parse_args` gets a new argument using a small named type-checker so bad values get argparse's existing usage-error treatment (consistent with how `--sort` already validates via `choices=`):

```
def positive_int(raw):
    value = int(raw)
    if value < 1:
        raise argparse.ArgumentTypeError("must be a positive integer")
    return value

parser.add_argument("--top", type=positive_int, default=None, metavar="N",
                    help="show only the top N campaigns by revenue per recipient")
```

- `build_report` gains one new optional parameter: `build_report(campaigns, sort_key="date", top=None)`. When `top` is `None` (the default), behavior and output are byte-identical to today. When set:

```
selected = campaigns
if top is not None:
    selected = sorted(campaigns, key=lambda c: -c.revenue_per_recipient)[:top]
ordered = sorted(selected, key=SORT_KEYS[sort_key])
...
```

So `--top` picks *which* campaigns qualify (by revenue per recipient, best first) and `--sort` still governs the *display order* of that subset - the two flags compose rather than one overriding the other. `--top` larger than the campaign count just returns everything (Python slicing handles this with no special-casing needed).

- `main()` passes `args.top` through to `build_report`.
- Whether `format_summary` recomputes over just the selected top-N campaigns or always reflects the full file is the one part of this feature I'm not certain about - see section 5, item 1. I'll implement whichever you confirm.
- Minor doc touch: I'll add a one-line `--top` example to the module docstring's usage block and to `README.md`'s usage/sort-keys section, since both currently document the CLI surface and would otherwise go stale. Small, proportionate addition, not a rewrite.

4. Tests to add

I'll add two new test files rather than editing `tests/test_campaign_report.py`, so the 5 baseline tests stay completely untouched and the new file's docstring purpose ("baseline, happy path only") stays accurate:

- **`tests/test_validation.py`** (uses `tmp_path` from pytest, already a dependency of the test runner - not a new project dependency):
  - missing file -> `CampaignReportError`/non-zero exit, message names the path
  - unreadable path via a directory (deterministic across platforms, avoids chmod flakiness) -> clear error
  - CSV missing a required column -> error names the missing column
  - non-numeric value in a numeric column -> error names row + column
  - negative value -> error names row + column
  - zero recipients -> error names the row/campaign
  - header-only CSV (zero data rows) -> clear, distinct error
  - completely empty (0-byte) file -> clear error
  - `main([bad_path])` returns 1 and stderr contains no `"Traceback"`, for at least one representative bad-input case
- **`tests/test_top_n.py`**:
  - `--top` limits the table to exactly N rows
  - selects the N highest `revenue_per_recipient` campaigns specifically (built from known in-memory `Campaign` fixtures, not the CSV, so expected ranking is unambiguous)
  - `--top` composes with `--sort`: same N rows, ordered by the requested sort key
  - `--top` greater than the campaign count returns all campaigns, no error
  - `--top 0` and `--top -1` are rejected with a usage error
  - whatever summary-scope behavior we confirm in section 5.1 gets its own explicit assertion

> **5. Ambiguities I want resolved before coding**
>
> 1. **Does the account-level summary recompute over just the top-N campaigns, or always reflect the whole file when `--top` is used?** The spec says `--top` composes with "the existing table **and summary** output," which reads to me like the summary should also narrow to the selected subset - but "summary always means the whole account, `--top` only trims the table" is an equally reasonable reading and changes the implementation. I'd lean toward recomputing over the subset, but want it confirmed.
> 2. **Should `--top`'s validation (rejecting 0/negative N) go through argparse's own mechanism (exit code 2, argparse's message format), or be funneled through the same `CampaignReportError` path as the CSV errors (exit code 1)?** I'm planning argparse-native, matching how `--sort` already works, but flagging the choice since the rest of the spec's error-handling language was written with CSV validation in mind.
> 3. **Scope of "unreadable" CSV path** - I'm planning to handle not-found, permission-denied, and is-a-directory as distinct messages under one `OSError` catch. Confirm that's sufficient and no other unreadable scenario is expected.
> 4. **Row-numbering convention in messages** - I'm planning 1-indexed data rows (first row under the header = "Row 1"), not raw file line numbers. Flagging in case you'd prefer literal file-line numbers instead.
> 5. **Zero recipients: hard failure vs. skip-and-continue?** I'm planning to treat it as a hard failure for the whole run (non-zero exit, no report produced), consistent with how the spec groups it alongside the other hard-failure classes. An alternative design would skip just that row and still report on the rest - materially different behavior, so flagging explicitly rather than assuming.
>
> Stopping here for your approval per Checkpoint 1. No files have been created, edited, or deleted.

Verified independently with `git status` on the supervising session: working tree clean, so the no-edits claim was true rather than merely asserted.

CP1 result and CP2 gate - Prompt

> CHECKPOINT 1 RESULT: plan approved, with three corrections and rulings on all five of your questions. Good plan. Asking about the summary scope instead of silently picking was the right call, and keeping the baseline test file untouched was a better instinct than I asked for.
>
> CORRECTION 1 - drop the blanket `except Exception` in main(). You flagged it yourself and you were right to. Catch CampaignReportError only. A bare `except Exception` satisfies the letter of "no raw tracebacks" while defeating the point of it: it would convert every genuine defect, including mine, into a polite one-line message and a silent exit 1. Unanticipated exceptions should crash loudly with a full traceback. I would rather see a real stack trace than a reassuring message that hides one.
>
> CORRECTION 2 - message wording. You plan to number 1-indexed data rows but write them as "Row 4". That reads as file line 4 and will send someone to the wrong line. Use the literal phrase "data row N".
>
> CORRECTION 3 - doc edits are approved but bounded. Add the --top example to the module docstring usage block and to README.md's usage/sort-keys section. Nothing else in README.md. Leave the existing docs/ reference alone.
>
> RULINGS ON YOUR QUESTIONS:

> 1. Summary scope: recompute over the selected top-N subset. A summary that aggregates campaigns not listed in the table above it cannot be reconciled by the person reading it.
> 2. --top validation: argparse-native, exit code 2, as you planned. A malformed flag is a usage error, not a data error.
> 3. Unreadable path: not-found, permission-denied, is-a-directory under one OSError catch is sufficient.
> 4. Row numbering: 1-indexed data rows, phrased per Correction 2.
> 5. Zero recipients: OVERRIDDEN. Do not fail the run. Skip that row, report the remaining campaigns normally, and print a warning to stderr naming the skipped campaign. Exit code stays 0 for an otherwise-valid file. Reasoning, so you can apply it to edge cases I have not listed: one malformed row should not deny an analyst the other seven campaigns, but a silent skip is worse than a crash, because a report that quietly describes less than the file contains is how a wrong number reaches a client deck. Continue, and say so loudly. A file where every row is skipped should be treated as having no usable campaigns.
>
> ADDITIONAL HARD RULE: do not edit tests/test_campaign_report.py. Not one line. If one of those 5 tests fails, that is a signal about your change, not a problem with the test.
>
> CHECKPOINT 2 - this is your next gate. Implement PART 1 ONLY (validation and error handling). Do NOT start Part 2. Do NOT implement --top at all, not even the argparse wiring.
>
> Work autonomously within Part 1: edit campaign_report.py, create tests/test_validation.py, run the tests, iterate until they pass. Do not commit anything.
>
> Stop when Part 1 is done and its tests pass, and report:
>
> 1. What you changed, file by file.
> 2. The result of `python3 -m pytest tests/ -q`.
> 3. Anything that surprised you or that you had to solve differently than planned.
> 4. Anything you are unsure about.
>
> Be honest in item 3 and 4 including about anything you got wrong on the way. I am recording this and a clean-looking report I later contradict with the diff is worse for you than an accurate one.

CP2 - Agent response (verbatim, 17 tool calls)

> Part 1 complete. All 33 tests pass. Report below.
>
> **1. What I changed, file by file**
>
> **`campaign_report.py`** (+126/-10)
>
> - Added `CampaignReportError(Exception)` - the single "bad input the user can fix" type. Docstring states explicitly that it is never for a defect in the module.
> - Added `NUMERIC_COLUMNS`, a tuple of `(column, cast)` pairs, replacing the five hardcoded `int(row[...])` / `float(row[...])` calls so parse-and-check happens in one loop instead of five places.
> - Added four helpers: `warn()` (stderr, `Warning:` prefix), `open_csv()` (`OSError` catch, distinct messages for not-found / is-a-directory / permission-denied / other), `check_columns()`, `parse_numeric()`.
> - Rewrote the body of `load_campaigns(path)`. Signature and return type unchanged (still a plain list, deliberately not a `(campaigns, warnings)` tuple, which would have broken baseline test 1).

- `main()` now catches `CampaignReportError` only, prints `Error: {message}` to stderr, returns 1. No blanket `except Exception`, per Correction 1.
- Did **not** touch `Campaign`, `totals`, `pct`, `money`, `format_row`, `format_header`, `format_summary`, `build_report`, or `parse_args`. No `--top` code of any kind, not even argparse wiring.

`tests/test_validation.py` (new, 28 tests) - grouped by the six error classes, plus an exit-code/stderr contract block and two valid-run guards.

`tests/expected_report.txt` (new, 1136 bytes) - **this file was not in my approved plan.** It is the golden copy of the valid-run output, captured from the unmodified script *before* I made any edit, and it backs one test asserting byte-for-byte identical output. Flagging it because you approved a plan that did not mention it. Say the word and I will delete it and the test that uses it.

`tests/test_campaign_report.py` - not touched. `git diff` confirms zero changes; all 5 pass unmodified.

Error messages produced (verified through the real CLI, not just in-process):

```
Error: CSV file not found: '.../nope.csv'
Error: '.../a_dir' is a directory, not a CSV file
Error: permission denied reading '.../locked.csv'
Error: '.../empty.csv' is empty: no header row found
Error: '.../header_only.csv' has no data rows: only a header was found
Error: '.../missing_cols.csv' is missing required column(s): revenue. Expected: campaign_id,
name, ...
Error: data row 1: column 'clicks' has a non-numeric value 'oops'
Error: data row 1: column 'revenue' is negative (-90.0); expected zero or more
Warning: data row 2: skipping campaign 'c2' (Dead Segment) because it has 0 recipients, so no
rate metric can be computed for it.
```

Zero-recipient row: skipped, warned, exit **0**, other campaigns reported. All-rows-skipped: warns per row, then errors with `has no usable campaigns: all 1 data row(s) were skipped`, exit 1.

2. `python3 -m pytest tests/ -q`

```
.................................                                        [100%]
33 passed in 0.03s
```

5 baseline + 28 new. Separately verified the output constraint by capturing all six valid invocations (default plus each of the five `--sort` keys) before editing and diffing after: **byte-identical, stderr empty, exit 0** in every case.

3. Surprises and things I did differently than planned

1. **I found a pre-existing crash I did not fix, and it is reachable with fully valid input.** A file whose campaigns sum to zero opens or zero orders (e.g. one row: `recipients=1000, opens=0, clicks=0, orders=0, revenue=0.00`) passes every check I added, then dies in `format_summary` with a raw `ZeroDivisionError` on `click_to_open_rate` (and `average_order_value` would do the same). I confirmed it pre-exists by running `git show HEAD:campaign_report.py` against the same file: identical traceback. So I did not introduce it, but it does mean Part 1's "no raw tracebacks" goal is not fully met. I deliberately left it because fixing it requires a display decision you have not made (print `n/a`, or hard-error), and you told me to stay inside Part 1. See item 4.
2. **Negative-value message shows the parsed number, not the raw text.** My plan said `is negative (-12.50)` using the original cell text; the code prints `(-90.0)` for a cell reading `-90.00`. Row and column

> are still named, so the analyst can still locate it, but the value is normalized rather than quoted verbatim.
>
> 3. **Helper functions are public, not underscore-prefixed as planned.** I wrote `warn`, `open_csv`, `check_columns`, `parse_numeric` instead of `_parse_numeric_field` etc., because every existing function in the module is public and adding a second naming convention to a 200-line script felt worse. This is still a deviation from what you approved, and it does grow the module's apparent API surface. Trivial to rename.
> 4. **`int()` rejects decimals, so `41250.5` in `recipients` reports as "non-numeric".** That acceptance behavior is inherited from the baseline (which also used `int()`), and I kept it so valid runs do not change. But my message wording calls a decimal "non-numeric", which is not quite true.
> 5. **A truncated row reports as a value problem, not a structural one.** `csv.DictReader` fills missing trailing fields with `None`, which my `TypeError` catch turns into `data row 1: column 'opens' has a non-numeric value (empty)`. Correct row and column, but it describes a short row as a bad cell.
> 6. **I did not make the `--top` doc edits** you approved in Correction 3. They document a flag that does not exist yet, so I held them for Part 2. Flagging in case you wanted them landed now.
>
> **4. What I am unsure about**
>
> 1. **The `ZeroDivisionError` above - needs your ruling.** Options: (a) display `n/a` for CTOR/AOV when the denominator is zero, which keeps the analyst's report per your Ruling 5 reasoning but invents a display convention you have not approved and touches `format_summary`; (b) raise `CampaignReportError` with a clear message, which is consistent with the rest of Part 1 but denies a report over data that is arguably legitimate; (c) leave it. I lean (a), one line, but it is your call, and I did not want to silently pick a formatting convention after you specifically praised asking rather than assuming.
> 2. **Keep or drop `tests/expected_report.txt`?** It is the strongest guard on "do not change the output of a valid run" and will keep guarding it through Part 2, but it is an unplanned file and a golden file is a maintenance object.
> 3. **Helper naming** (item 3.3) - public or underscore-private?
> 4. **A UTF-8 BOM in the export would report `campaign_id` as a missing column** rather than naming the BOM. The baseline would `KeyError` on the same file, so this is still an improvement, and switching to `encoding="utf-8-sig"` would fix it properly. I did not change the read encoding because that alters the parse path and was not in scope.
> 5. **Message-wording fixes for items 3.4 and 3.5** are cheap if you want them, but both would make messages longer, and I did not want to churn wording you already corrected once.
>
> Nothing committed. Working tree holds one modified file and two new files, all on `feature/validation-and-top-n`.

Supervisor verification of this report is tabulated in `docs/02-checkpoint-records.md`, Checkpoint 2. All four checkable claims held, including the claim that assigned the `ZeroDivisionError` to me rather than to itself.

CP3 - Review with changes requested (prompt, abridged to the requests)

The full review prose is in `docs/03-code-review.md`. The six requests as issued:

> CHECKPOINT 3 REVIEW: CHANGES REQUESTED. Not a rejection. [...] My own held-out acceptance suite, which you have never seen and which was committed before you got the task, passes 9 of 10 against your code. The one failure, AC8, is MY bug, not yours. AC8 demanded --top 3 print the three campaigns in descending rpr order. At CP1 I ruled that --top selects and --sort orders. You implemented my ruling exactly. I never propagated that ruling into AC8, in the same edit where I amended AC6. Your output is correct; my acceptance test was stale.
>
> I ALSO FOUND ONE YOU MISSED: build_report(campaigns, top=0) raises ZeroDivisionError. positive_int guards the CLI, but top_campaigns and build_report are public and unguarded [...] Validation living only at the CLI boundary is fine until someone imports the module, which is exactly what your own tests do.
>
> I also reproduced your SORT_KEYS concern concretely, and it is worse than you described. Reassigning SORT_KEYS["rpr"] to ascending, a plausible display-only change, makes top_campaigns return the three WORST campaigns [...] Silently.
>
> REQUIRED CHANGES. I am explicitly unfencing format_summary and the formatting layer for this round. That fence existed to prevent gratuitous rewriting, not to force you to ship a crash.
>
> 1. Fix the zero-denominator crash, properly. Render "n/a" [...] Your CP2 option (a), which I wrongly declined. My reasoning has changed because the facts changed: this is no longer a pre-existing defect on unrelated input, it is a defect your feature makes reachable on input that worked before.
> 2. Decouple selection from display [...] Add a test that would fail if someone flipped the display lambda.
> 3. Guard the empty/zero selection path [...]
> 4. NUMERIC_KINDS[cast] -> use a .get fallback. An error reporter that raises KeyError while reporting an error is the worst possible failure site.
> 5. Drop the TypeError from positive_int.
> 6. Fix "1 campaigns".
>
> Do NOT do anything else. [...] If byte-identity breaks for the 8-campaign fixture, stop and tell me rather than adjusting anything to compensate.
>
> [...] Also tell me if any of my six requests was wrong.

CP3 - Agent remediation response (verbatim excerpts)

> All six changes are in. 97 tests pass, byte-identity holds.

On the guard, item 3:

> I chose **raising** `ValueError` over returning an empty report, in both `top_campaigns` (`count < 1`) and `build_report` (empty `campaigns`). `ValueError`, not `CampaignReportError`, deliberately: `CampaignReportError` is documented as user-fixable input and `main()` converts it to a polite exit 1, but a caller passing `top=0` is a programming error and should not be dressed up as bad user input.

Disclosing that it changed two of its own tests:

> **Two of my own tests failed and I changed them, not the code:**
> `test_no_campaign_is_summarized_without_being_listed` and

> `test_top_composes_with_a_skipped_zero_recipient_row` both asserted the literal `"1 campaigns"`. Item 6 is precisely what made them wrong, so I updated them to expect correct pluralization.

The zero-orders file with `--top 1`, which previously produced a traceback:

```
CAMPAIGN                   SENT        RECIPIENTS    OPEN   CLICK     CVR      REVENUE
RPR
-------------------------------------------------------------------------------------------
-----
Attribution Lag            2026-03-04      1,000   50.0%    5.0%    0.0%      $900.00
$0.90

1 campaign  |  1,000 recipients  |  $900.00 revenue
Blended open 50.0%  |  click 5.0%  |  CTOR 10.0%  |  CVR 0.0%
Revenue per recipient $0.90  |  AOV n/a
```

> Only the genuinely undefined metric degrades. CTOR, RPR, and revenue stay real numbers, and `CVR 0.0%` is correct rather than `n/a` because zero orders over 1,000 recipients is a defined rate of zero.

Answering "was any of my six requests wrong?", the correction to my reasoning:

> **Item 3's rationale is partly superseded by item 1, though the change is still worth making.** You justified it as "so `build_report` and `top_campaigns` cannot be made to divide by zero by a caller." I tested this by applying item 1 and then removing the item 3 guards: `build_report(campaigns, top=0)` does **not** divide by zero any more. [...] So item 1 already defuses the crash. What item 3 actually prevents is a *meaningless report* rather than an exception. I implemented it as you asked and I think raising is right, but the guard earns its place as "refuse to describe nothing", not "prevent a ZeroDivisionError".

And on its own invalid first verification of item 4:

> I initially tested with `Decimal`, which proved nothing: `decimal.InvalidOperation` is an `ArithmeticError`, so it escapes `except (TypeError, ValueError)` before the message is ever built. Retested with an unmapped cast that raises `ValueError` and got the correct fallback.

Supervisor verification of this round is recorded in `docs/02-checkpoint-records.md`, Checkpoint 3. I re-derived the byte-identity claim, the 97-test count, the untouched baseline suite, the fixed crash, and the correction to my own item-3 rationale.

CPSC 6820 Literature Component

The +25 graduate requirement for Assignment 4.1. Jeff Branyon. Summer 2026.

Included as the final part of this document rather than as a separate upload, because only one Canvas item was open for Week 4. Everything before this page is Assignment 4.1. This part is self-contained and can be graded on its own.

Paper

Hussein Mozannar, Gagan Bansal, Adam Fourney, and Eric Horvitz. 2024. Reading Between the Lines: Modeling User Behavior and Costs in AI-Assisted Programming. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (CHI '24), May 11-16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, Article 142, 16 pages. https://doi.org/10.1145/3613904.3641936

Summary and connection

Methodology. Mozannar et al. ran a 60-minute observational study of 21 programmers using GitHub Copilot in VS Code on a remote VM. Familiarity ranged from 11 never-users to 7 weekly users. Each received one of eight mostly-Python tasks in a 20-minute block, delivered as an image so participants could not paste the prompt and had to author their own. Telemetry logged every show, accept, reject, and browse event, segmenting each session. Immediately afterward, participants reviewed their own screen recording segment by segment and labeled each with one of twelve CUPS states (CodeRec User Programming States). That retrospective self-labeling is the methodological core: rather than inferring intent from telemetry, it asks developers what they were doing. Analysis covered 3,137 labels; 353 uncertain ones were discarded, not reinterpreted. There was deliberately no Copilot-free control: the aim was time allocation within AI-assisted work, not speedup.

Findings. Verifying suggestions, an activity the tool newly introduced, was the largest single one at 22.4% of session time, ahead of writing new functionality at 14.1%. Copilot states consumed about 51% of sessions. The transition graph shows that deferring thought on a suggestion leads to verifying it later with probability 0.54, a pattern they name **verification debt**: deference postpones review rather than removing it. Most consequentially, counting verification that happens *after* acceptance raises mean verification time from 3.25 to 15.21 seconds, roughly fivefold. Acceptance-rate metrics therefore understate review cost systematically.

Connection to my lab. Verification debt is the empirical case for my Checkpoint 2, and I did not have that argument when I designed it. My instinct was that two gates would do: plan approval, then a final diff review. The fivefold correction argues otherwise, because one end-of-task review must absorb debt accrued across the entire run, which is exactly what a mid-stream gate prevents. My lab reproduced the cost asymmetry at small scale. The agent produced Part 1 in roughly five minutes of wall clock; verifying four of its claims against the baseline commit, reading a 126-line diff line by line, and auditing 23 agent-

written tests for gaming is the part that does not compress, and it took nearly all of my own effort. Delegation did not remove work, it converted authoring into verifying. That reframes what a checkpoint is worth: not that it prevents a bad merge, but that it divides an unreviewable quantity of verification into amounts a human will actually perform.